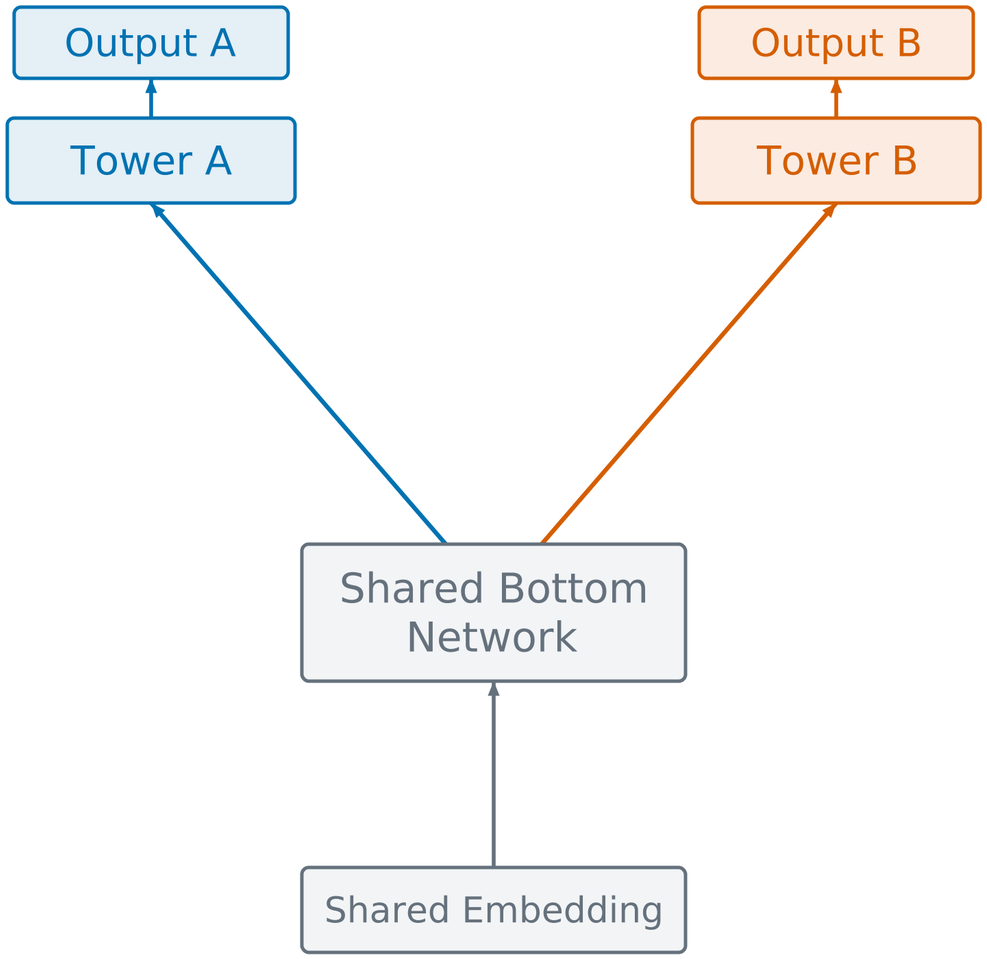
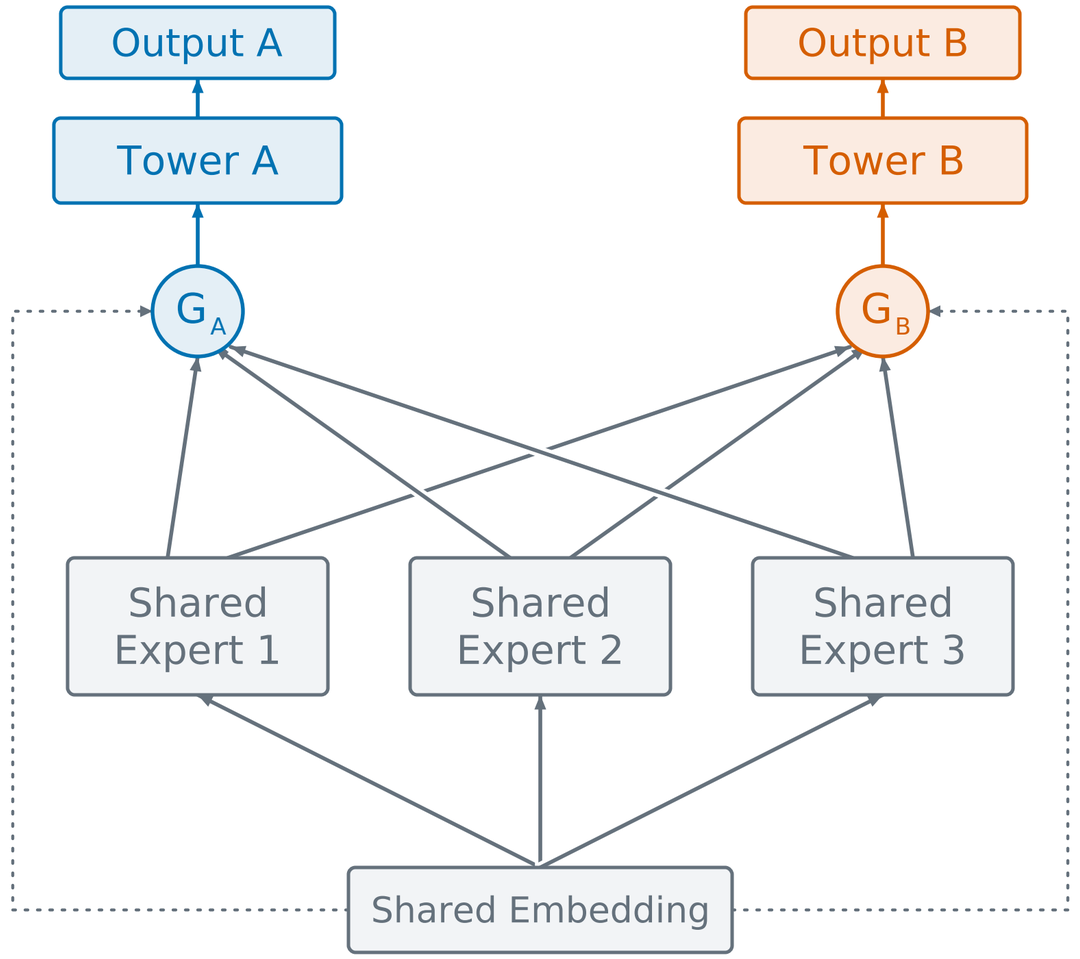
(a) Shared-Bottom



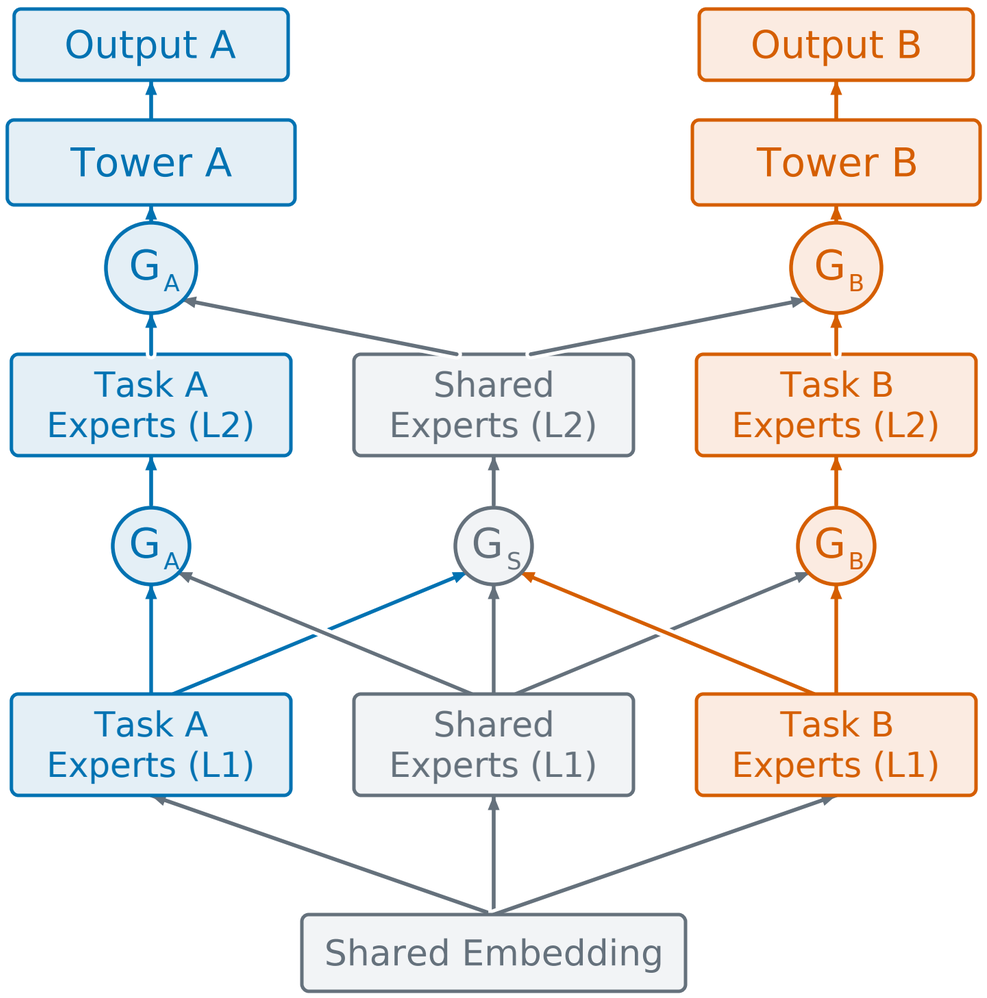
Task-specific towers; shared representation

(b) MMoE



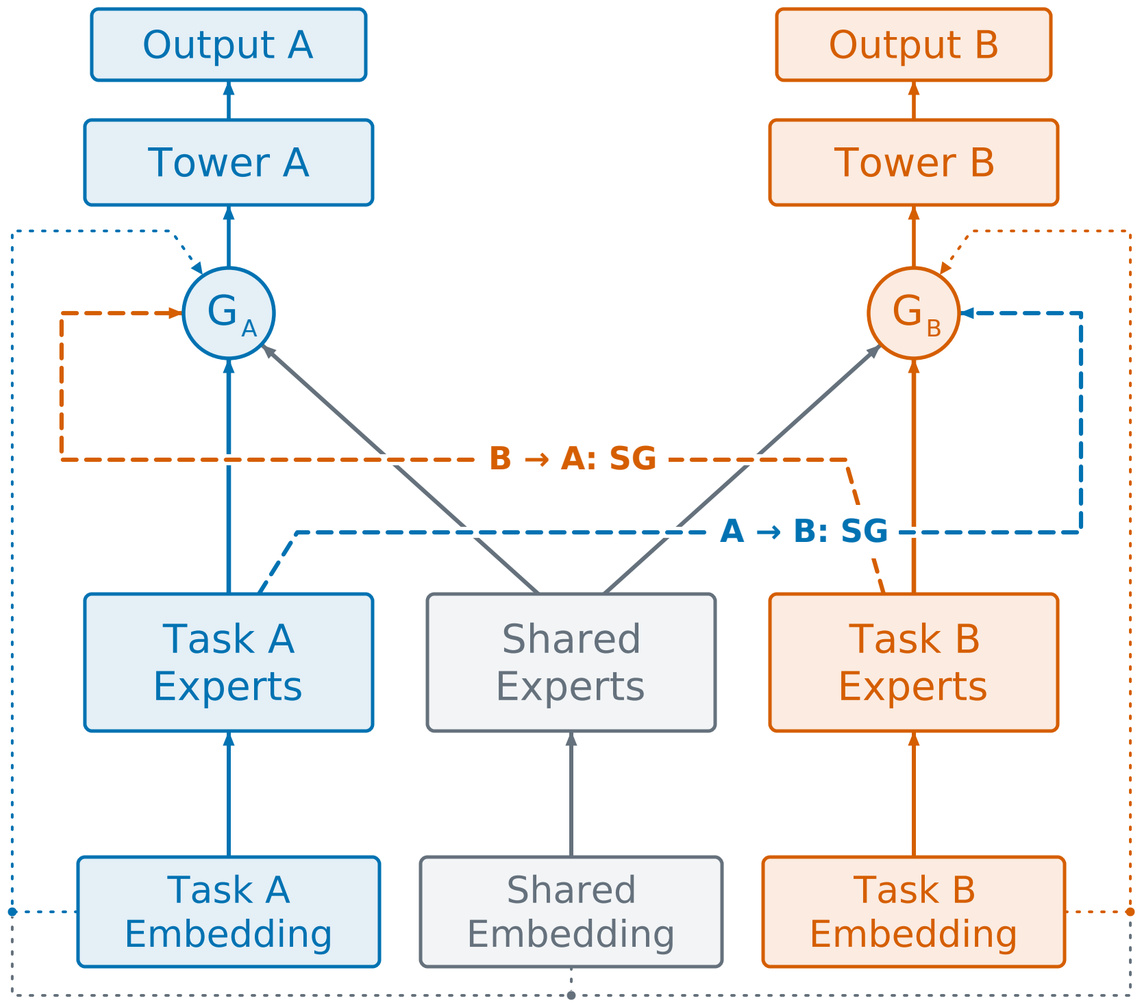
Shared experts; task-specific, input-dependent gates

(c) PLE



Two extraction levels; one box per expert group

(d) STEM-Net



Gate inputs: shared + corresponding task embedding

Task A Task B Shared G: gate + weighted aggregation

Representation flow Gate conditioning SG: forward pass; stop gradient

Crossings without a dot are not connections. Expert boxes in PLE/STEM-Net denote groups.